

RAI 818 – INTELLIGENT MOBILE ROBOTICS

Extended Kalman Filter

- **Process Noise Covariance**
- **Extended Kalman Filter (EKF)**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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Kalman Filter

- Kalman Filter (Revised)

Kalman Filter Equations - Revised

Prediction

$$\vec{x}_t = F_t \vec{x}_{t-1} + B_t u_t$$

$$\vec{P}_t = F_t P_{t-1} F_t^T + Q$$

Correction

$$S_t = H_t P_t H_t^T + R$$

$$K_t = P_t H_t^T S_t^{-1} \Rightarrow K_t = P_t H_t^T (H_t P_t H_t^T + R)^{-1}$$

$$x_t = \vec{x}_t + K_t \vec{y}_t \quad \vec{y}_t = z_t - H_t \vec{x}_t \Rightarrow x_t = \vec{x}_t + K_t (z_t - H_t \vec{x}_t)$$

$$P_t = \vec{P}_t - K_t H_t \vec{P}_t$$

Prediction

$$\vec{x}_t = F_t \vec{x}_{t-1} + B_t u_t$$

$$\vec{P}_t = F_t P_{t-1} F_t^T + Q$$

Correction

$$K_t = P_t H_t^T (H_t P_t H_t^T + R)^{-1}$$

$$x_t = \vec{x}_t + K_t (z_t - H_t \vec{x}_t)$$

$$P_t = \vec{P}_t - K_t H_t \vec{P}_t$$



Kalman Filter

- Kalman Filter (Revised)
- Process Noise Covariance

Process Noise Covariance

$$P_t = F_t P_{t-1} F_t^T + Q_t$$

$$\vec{x}_k = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad Q = \begin{bmatrix} q_{11} & 0 & \dots & 0 \\ 0 & q_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & q_{nn} \end{bmatrix}$$

$$P_t = F_t P_{t-1} F_t^T + F_t Q F_t^T$$



Kalman Filter

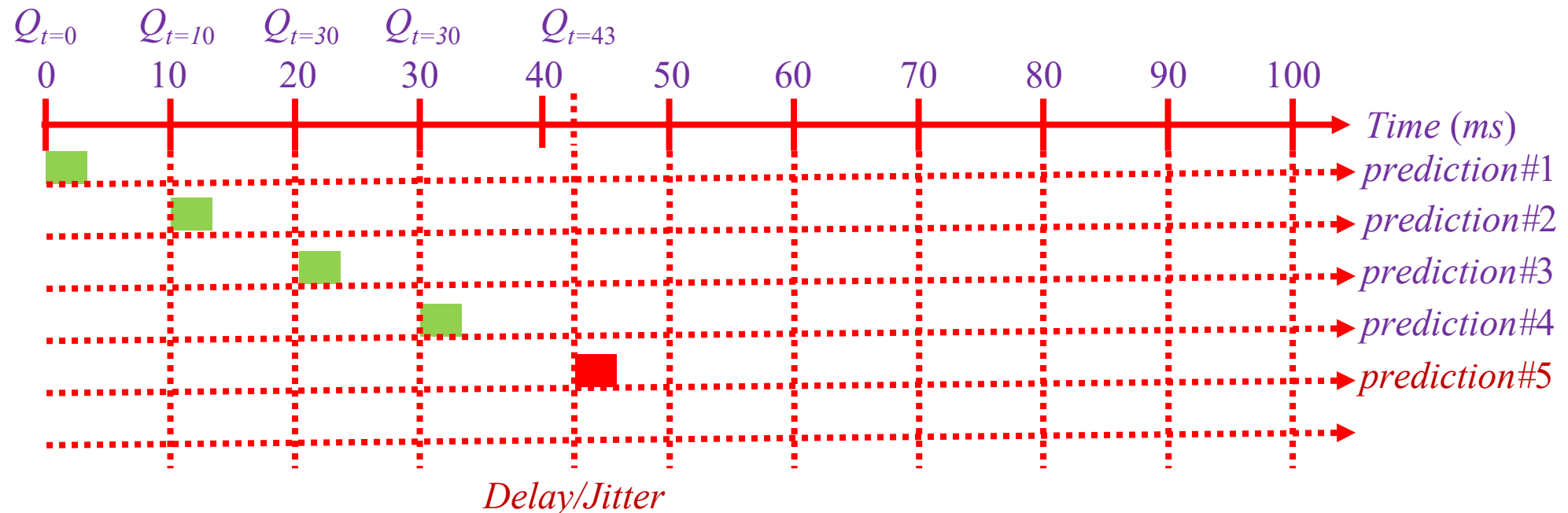
- Kalman Filter (Revised)
- Process Noise Covariance

Process Noise Covariance

$$P_t = F_t P_{t-1} F_t^T + F_t Q_t F_t^T$$

$$P_t = F_t P_{t-1} F_t^T + \int_{t=0}^{t=1} F_t Q F_t^T dt$$

$$\int_{t=30}^{t=43} Q_t dt$$



Defining futures

Kalman Filter

- Kalman Filter (Revised)
- Process Noise Covariance
 - Example

Process Noise Covariance - Example

$$p_k = p_{k-1} + tv_{k-1} \quad \begin{bmatrix} p_k \\ v_k \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} p_{k-1} \\ v_{k-1} \end{bmatrix} \quad Q = \begin{bmatrix} q_{pp} & 0 \\ 0 & q_{vv} \end{bmatrix}$$

System Model

$$\begin{aligned} Q'_t &= \int_{t_0}^{t_1} F_t Q F_t^T dt = \int_{t_0}^{t_1} \begin{bmatrix} q_{pp} + q_{vv} t^2 & q_{vv} t \\ q_{vv} t & q_{vv} \end{bmatrix} dt \\ &= \left[\begin{bmatrix} q_{pp} t + q_{vv} t^3 / 3 & q_{vv} t^2 / 2 \\ q_{vv} t^2 / 2 & q_{vv} t \end{bmatrix} \right]_{t_0}^{t_1} = \begin{bmatrix} q_{pp} (t_1 - t_0) + q_{vv} (t_1 - t_0)^3 / 3 & q_{vv} (t_1 - t_0)^2 / 2 \\ q_{vv} (t_1 - t_0)^2 / 2 & q_{vv} (t_1 - t_0) \end{bmatrix} \\ &= \begin{bmatrix} q_{pp} (\Delta t) + q_{vv} (\Delta t)^3 / 3 & q_{vv} (\Delta t)^2 / 2 \\ q_{vv} (\Delta t)^2 / 2 & q_{vv} (\Delta t) \end{bmatrix} \end{aligned}$$

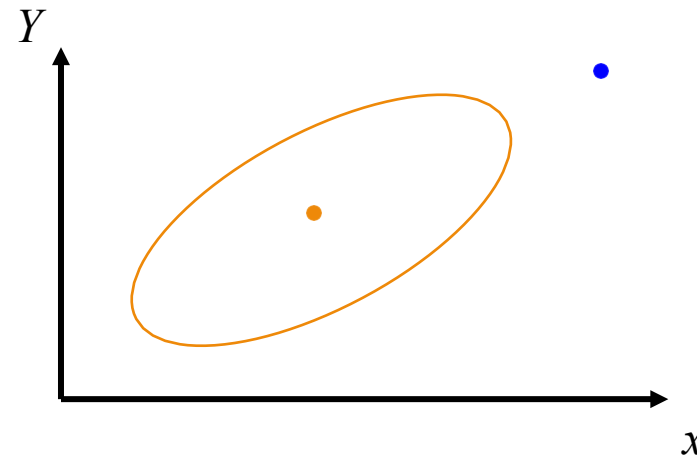
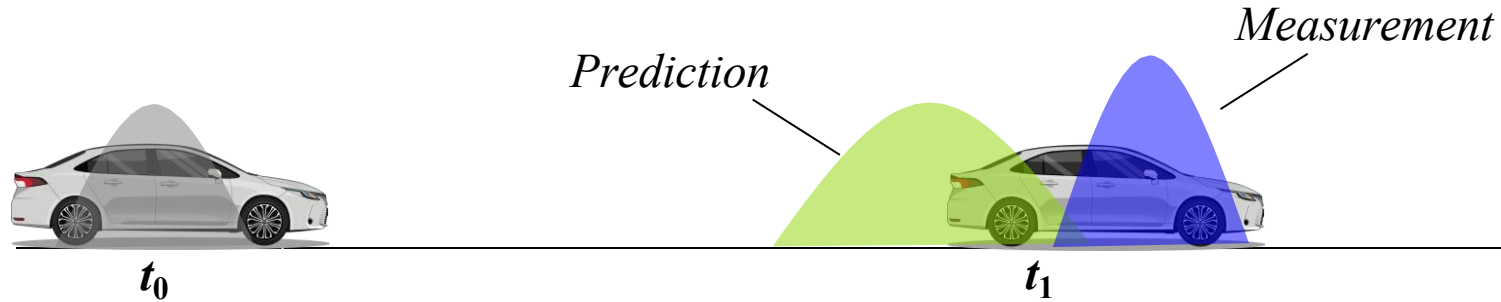


Kalman Filter



- Kalman Filter (Revised)
- Process Noise Covariance
 - Example

Process Noise Covariance



Under-Estimated Covariance



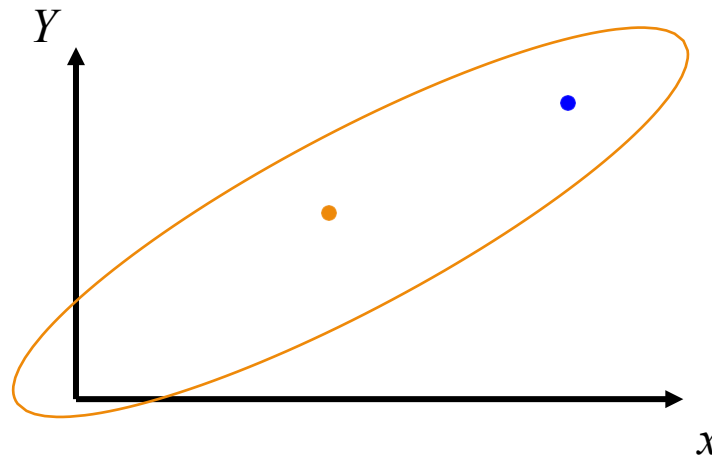
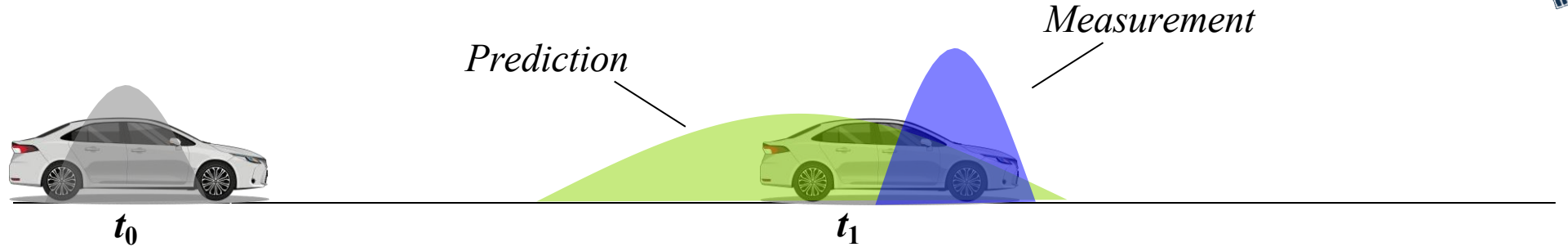
Defining futures

Kalman Filter



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Process Noise Covariance



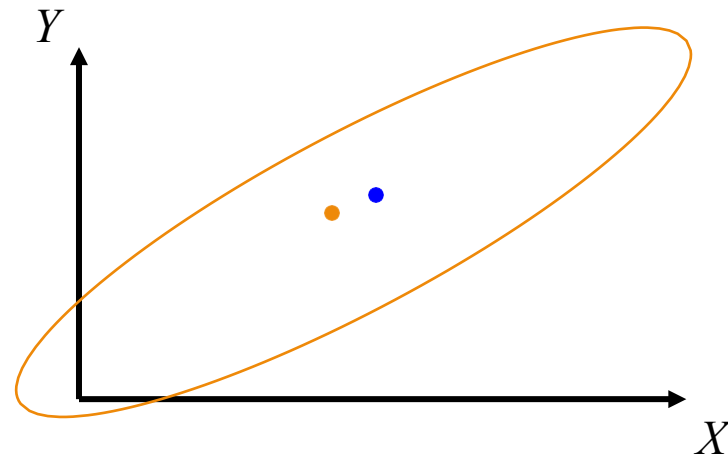
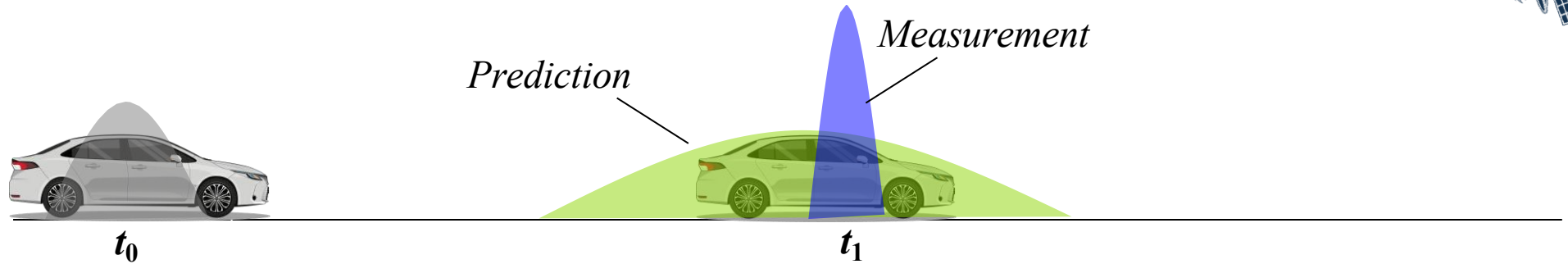
Defining futures

Kalman Filter



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Process Noise Covariance



Over-Estimated Covariance



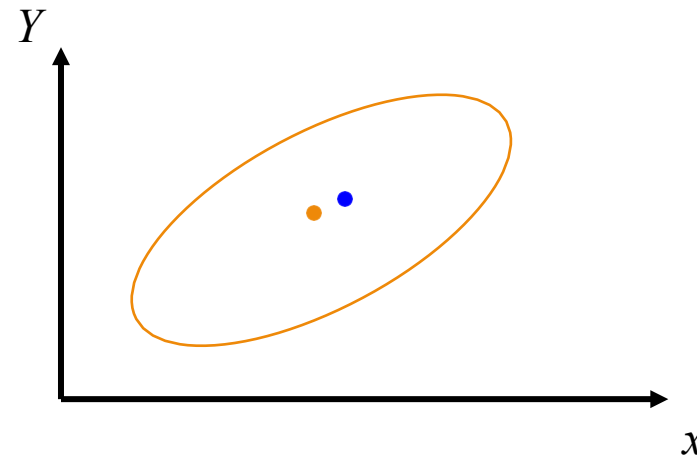
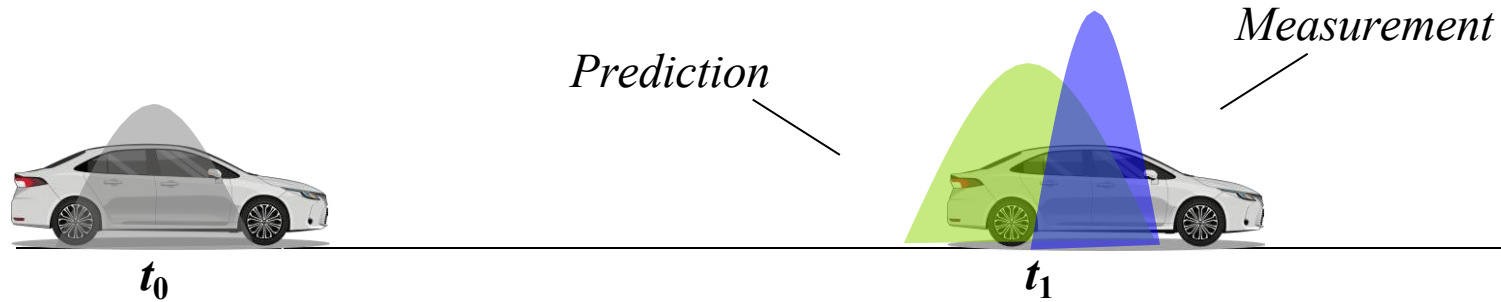
Defining futures

Kalman Filter



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Process Noise Covariance



Defining futures

Kalman Filter

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 - Example
 - Mahalanobis Distance

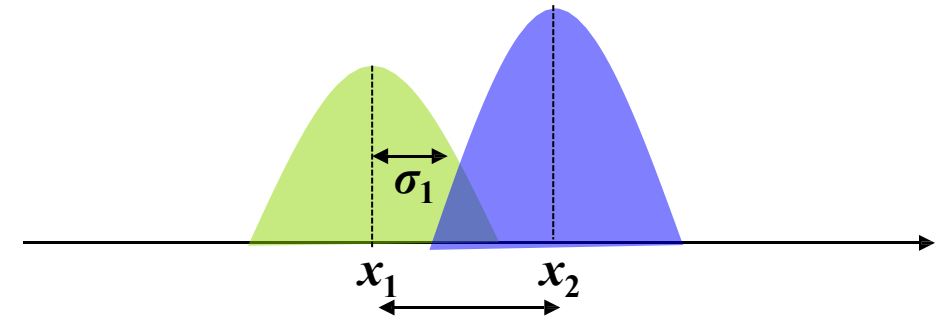
Mahalanobis Distance

$$(\Delta x')^2 = \frac{(x_2 - x_1)^2}{\sigma_1^2}$$

$$\begin{matrix} d^2 & = & (\vec{x}_2 - \vec{x}_1)^T & \Sigma^{-1} & (\vec{x}_2 - \vec{x}_1) \\ \begin{matrix} 1 \times 1 \end{matrix} & & \begin{matrix} 1 \times n \end{matrix} & \begin{matrix} n \times n \end{matrix} & \begin{matrix} 1 \times n \end{matrix} \end{matrix}$$

$$d^2 = (\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1)$$

$$d = \sqrt{(\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1)}$$



← Normalized Innovation Square Distance

← Mahalanobis Distance

1. Run the System
2. At each measurement update step calculate the Mahalanobis Distance d between the received measurement and the predicted state using the above formula.
3. Compare the calculated distance d against a gating threshold. If this is less than the threshold, the measurement is inside the uncertainty estimated for the predicted state. Otherwise, it lies outside the uncertainty



Kalman Filter

- Bayes Theorem
- Bayesian Inference
- Example
- Kalman Filter
- **Example**

Process Noise Covariance

$$d^2 = (\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1) \longleftarrow \text{Normalized Innovation Square Distance}$$

$$d = \sqrt{(\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1)} \longleftarrow \text{Mahalanobis Distance}$$

99.95 % of the samples lie
inside the uncertainty range

	P										
DF	0.995	0.975	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	0.0000393	0.000982	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.550	10.828
2	0.0100	0.0506	3.219	4.605	5.991	7.378	7.824	9.210	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.860	16.924	18.467
5	0.412	0.831	7.289	9.236	11.070	12.833	13.388	15.086	16.750	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.549	20.701	22.459



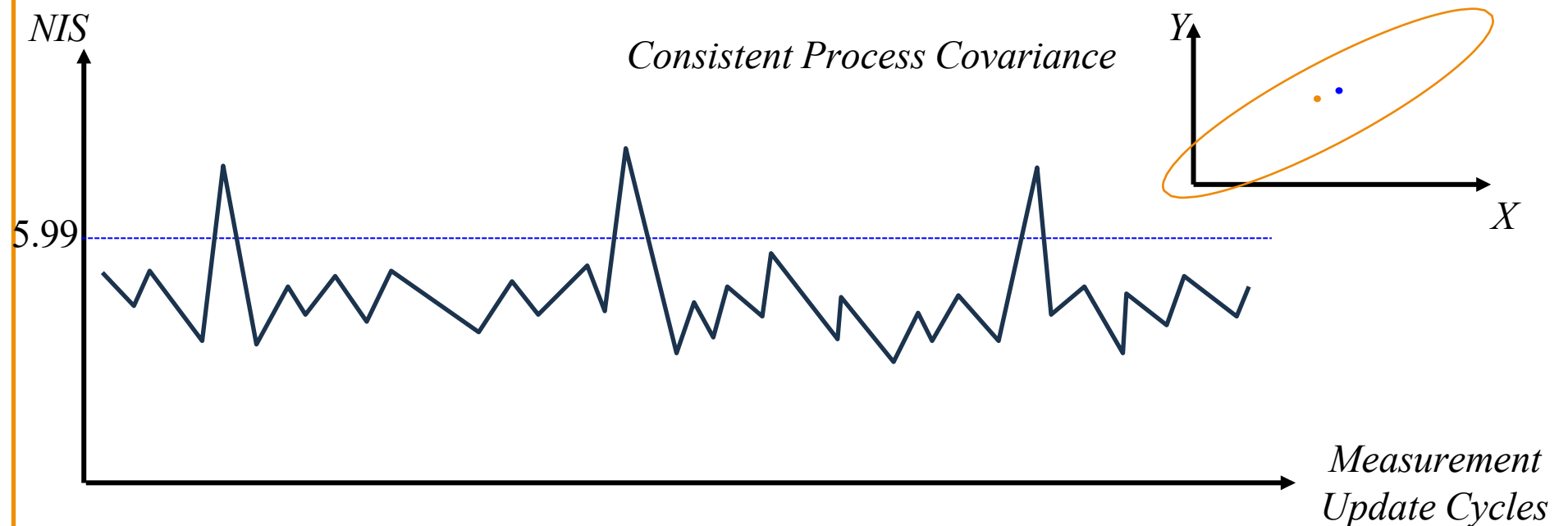
Kalman Filter

- Kalman Filter (Revised)
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 - Example
 - Mahalanobis Distance

Mahalanobis Distance

$$d^2 = (\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1) \longleftarrow \text{Normalized Innovation Square Distance}$$

$$d = \sqrt{(\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1)} \longleftarrow \text{Mahalanobis Distance}$$



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Mahalanobis Distance

$$d^2 = (\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1) \longleftarrow \text{Normalized Innovation Square Distance}$$

$$d = \sqrt{(\vec{x}_2 - \vec{x}_1)^T \Sigma^{-1} (\vec{x}_2 - \vec{x}_1)} \longleftarrow \text{Mahalanobis Distance}$$

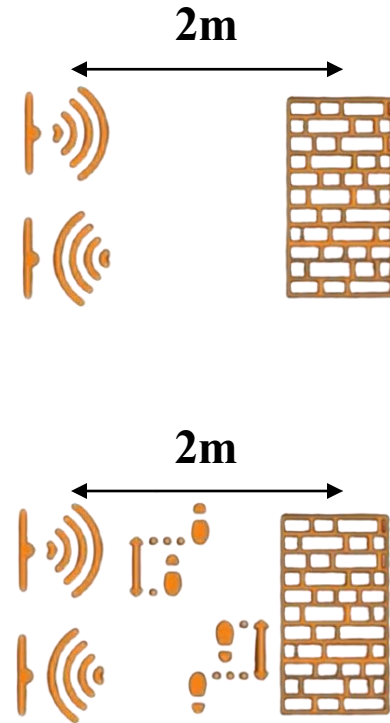
1. Measurement noise covariance can (in some cases) be known from the sensor datasheet directly
2. Most of the time it needs to be found statistically by gathering actual sensor measurements in a statically controlled environment and comparing them against ground-truth/actual data



Kalman Filter

- Kalman Filter (Revised)
- Process Noise Covariance
 - Example
 - Mahalanobis Distance
 - Sensor Example

Process Noise Covariance – Sensor Example



$$\bar{x} = \frac{1}{n} \sum_{i=0}^n x_i \quad P = \frac{1}{n} \sum_{i=0}^n (x_i - \bar{x})(x_i - \bar{x})^T$$



Defining futures

Kalman Filter

- Kalman Filter (Revised)
- Process Noise Covariance
- Extended Kalman Filter (EKF) – Non-Linear Case

Extended Kalman Filter – Non Linear Case

Prediction

~~$$\vec{x}_t = F_t \vec{x}_{t-1} + B_t u_t$$~~

$$\vec{P}_t = F_t P_{t-1} F_t^T + Q$$

$$\vec{x}_t = f(\vec{x}_{t-1}, u_t)$$

Correction

$$S_t = H_t P_t H_t^T + R$$

$$K_t = P_t H_t^T (H_t P_t H_t^T + R)^{-1}$$

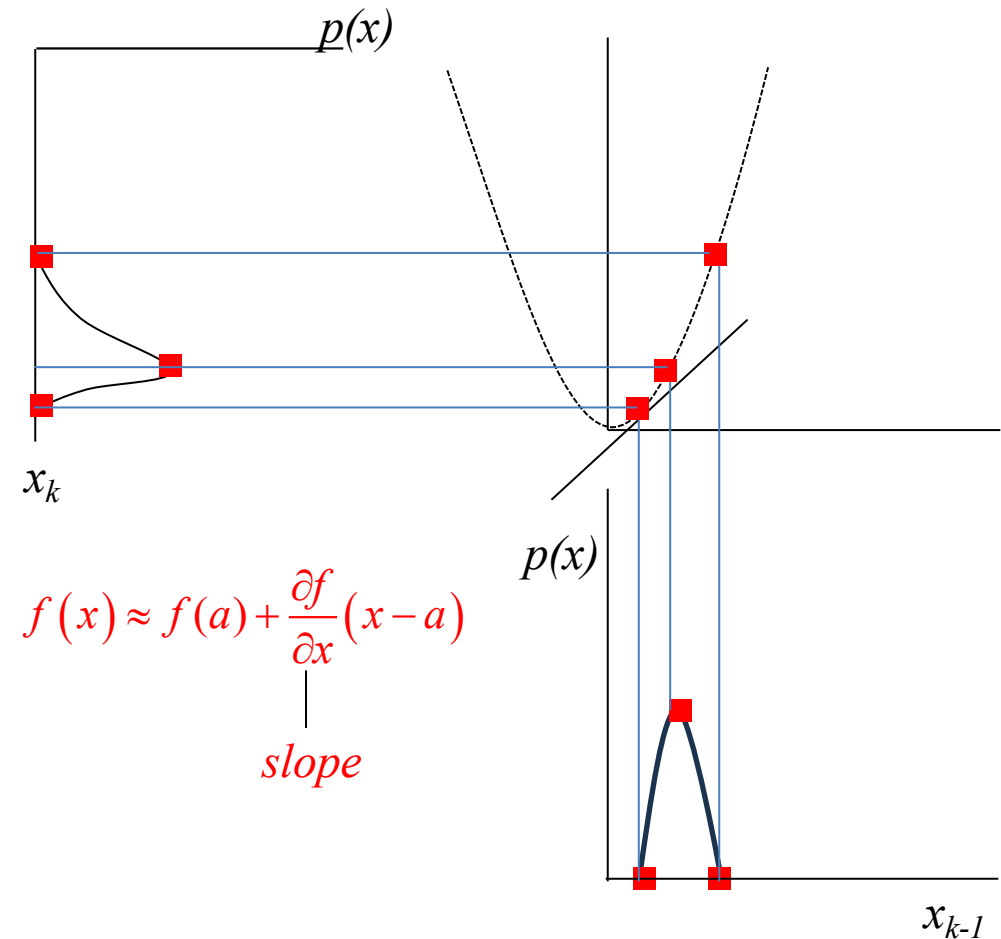
~~$$x_t = \vec{x}_t + K_t (z_t - H_t \vec{x}_t)$$~~

$$x_t = \vec{x}_t + K_t (z_t - h(\vec{x}_t))$$

$$P_t = \vec{P}_t - K_t H_t \vec{P}_t$$

~~$$y = mx + e$$~~

$$y = x^2$$



Kalman Filter

- Kalman Filter (Revised)
- Process Noise Covariance
- Extended Kalman Filter (EKF) – Non-Linear Case

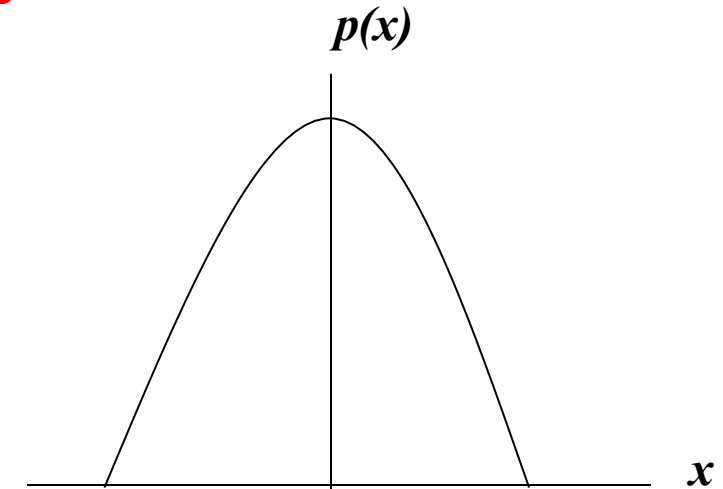
Extended Kalman Filter – Non Linear Case

$$f(x) \approx f(a) + \frac{\partial f}{\partial x}(x-a)$$

|
Slope/Jacobian

$$F_t = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{bmatrix}$$

*Jacobian Matrix
for multivariate
system (3D)*



1 dimension: $\left(\frac{\partial f}{\partial x}\right)^2 E[x]^2 = \left(\frac{\partial f}{\partial x}\right) \sigma^2$

n dimension: $F_t E[\vec{x}\vec{x}^T] F_t^T = F_t P F_t^T$



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Extended Kalman Filter – Non-Linear Case

Prediction

$$\vec{x}_t = f(\vec{x}_{t-1}, u_t)$$

$$\vec{P}_t = F_t P_{t-1} F_t^T + B_t U_t B_t^T + Q$$

Correction

$$S_t = H_t P_t H_t^T + R$$

$$K_t = P_t H_t^T S_t^{-1}$$

$$x_t = \vec{x}_t + K_t (z_t - h(\vec{x}_t))$$

$$P_t = \vec{P}_t - K_t H_t \vec{P}_t$$

$$F_t = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{bmatrix}$$

$$B_t = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \\ \frac{\partial f_3}{\partial u_1} & \frac{\partial f_3}{\partial u_2} \end{bmatrix}$$

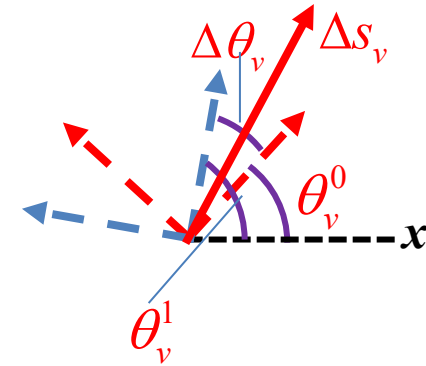
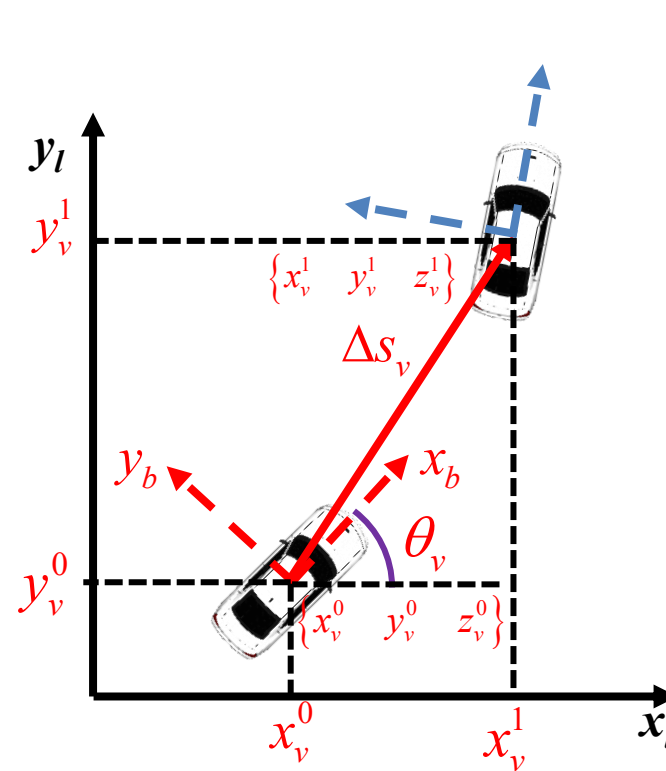
$$H_t = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \frac{\partial h_1}{\partial x_3} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \frac{\partial h_2}{\partial x_3} \end{bmatrix}$$



Kalman Filter

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Extended Kalman Filter – Non Linear Case



$$\Delta s = v \cdot \Delta t$$

$$\Delta \theta = \dot{\theta} \cdot \Delta t$$

$$\theta_v^1 = \theta_v^0 + \Delta \theta_v$$

$$x^1 = x^0 + \Delta s^1 \cos \left(\theta_v^0 + \frac{\Delta \theta^1}{2} \right)$$

$$y^1 = y^0 + \Delta s^1 \sin \left(\theta_v^0 + \frac{\Delta \theta^1}{2} \right)$$



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Prediction

$$\vec{x}_t = f(\vec{x}_{t-1}, u_t)$$

$$\vec{P}_t = F_t P_{t-1} F_t^T + B_t U_t B_t^T + Q$$

$$\vec{x}_t = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \quad \vec{u}_t = \begin{bmatrix} \Delta s \\ \Delta \dot{\theta} \end{bmatrix}$$

$$\vec{x}_t = f(\vec{x}_{t-1}, u_t) = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} x^1 = x^0 + \Delta s^1 \cos\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) \\ y^1 = y^0 + \Delta s^1 \sin\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) \\ \theta_v^1 = \theta_v^0 + \Delta \theta_v \end{bmatrix}$$

$$F_t = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\Delta s^1 \sin\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) \\ 0 & 1 & \Delta s^1 \cos\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) \\ 1 & 0 & 1 \end{bmatrix}$$

$$B_t = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \\ \frac{\partial f_3}{\partial u_1} & \frac{\partial f_3}{\partial u_2} \end{bmatrix} = \begin{bmatrix} \cos\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) & -\frac{1}{2} \Delta s \sin\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) \\ \sin\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) & \frac{1}{2} \Delta s \cos\left(\theta_v^0 + \frac{\Delta \theta^1}{2}\right) \\ 0 & 1 \end{bmatrix}$$



Defining futures